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decrease in the intrapulmonary pressure forces the fresh air from outside to enter the air passages into the lungs to equalize the pressure. This process is called **inspiration.**

Relaxation of the diaphragm allows the diaphragm and sternum to return to its dome shape and the internal intercostal muscles contract, pulling the ribs downward reducing the thoracic volume and pulmonary volume. This results in an increase in the intrapulmonary pressure slightly above the atmospheric pressure causing the expulsion of air from the lungs. This process is called **expiration.**

On an average, a healthy human breathes 12–16 times/minute. An instrument called Spirometer is used to measure the volume of air involved in breathing movements for clinical assessment of a person's pulmonary function.

### 6.3.1 Respiratory volumes and capacities

The volume of air present in various phases of respiration is denoted as

**Respiratory volumes:** (Figure 6.5)

- **Tidal Volume (TV)** Tidal volume is the amount of air inspired or expired with each normal breath. It is approximately 500 mL., i.e. a normal human adult can inspire or expire approximately 6000 to 8000mL of air per minute. During vigorous exercise, the tidal volume is about 4–10 times higher.

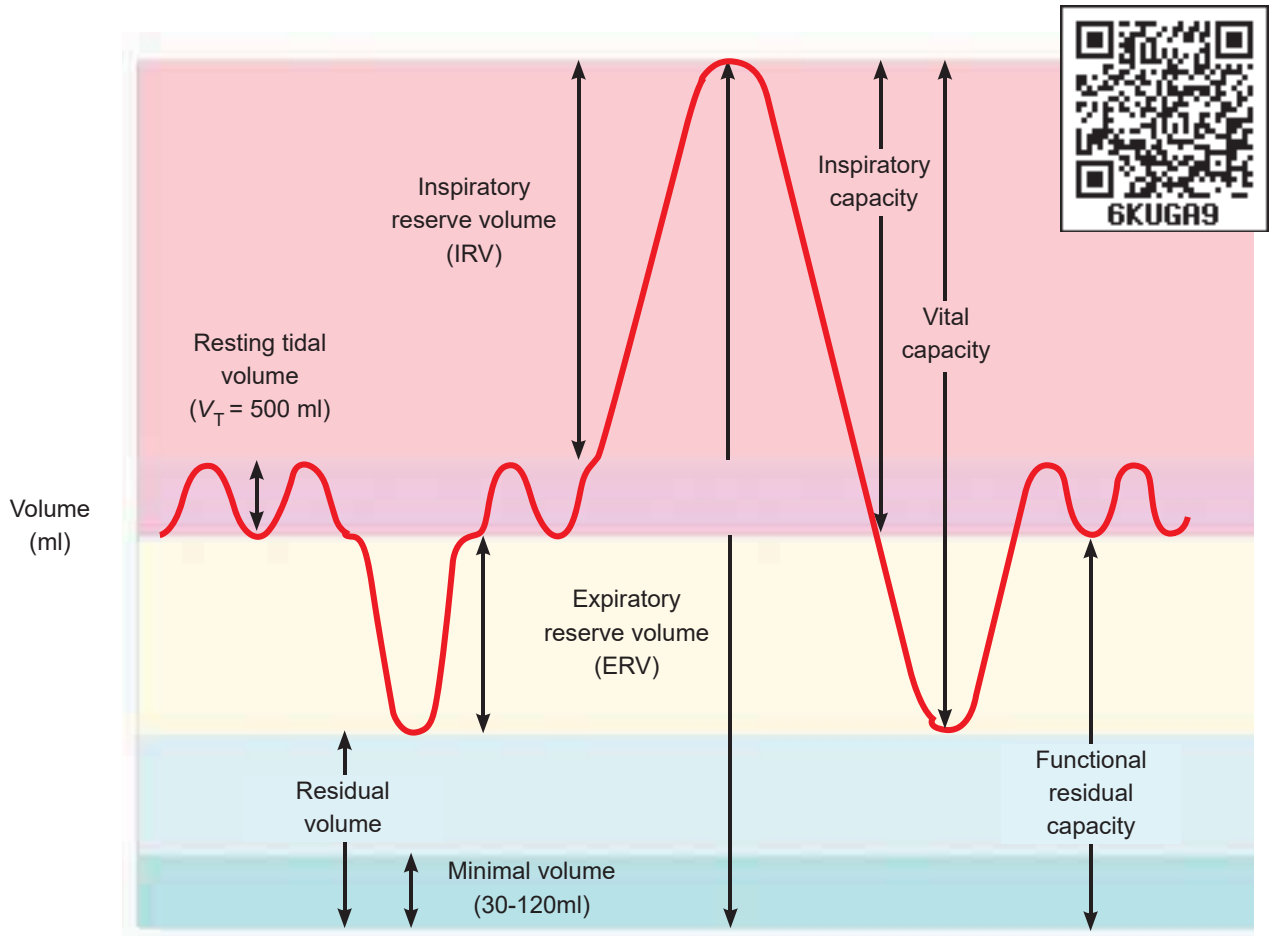



**Figure 6.5 Lung volumes and capacity**